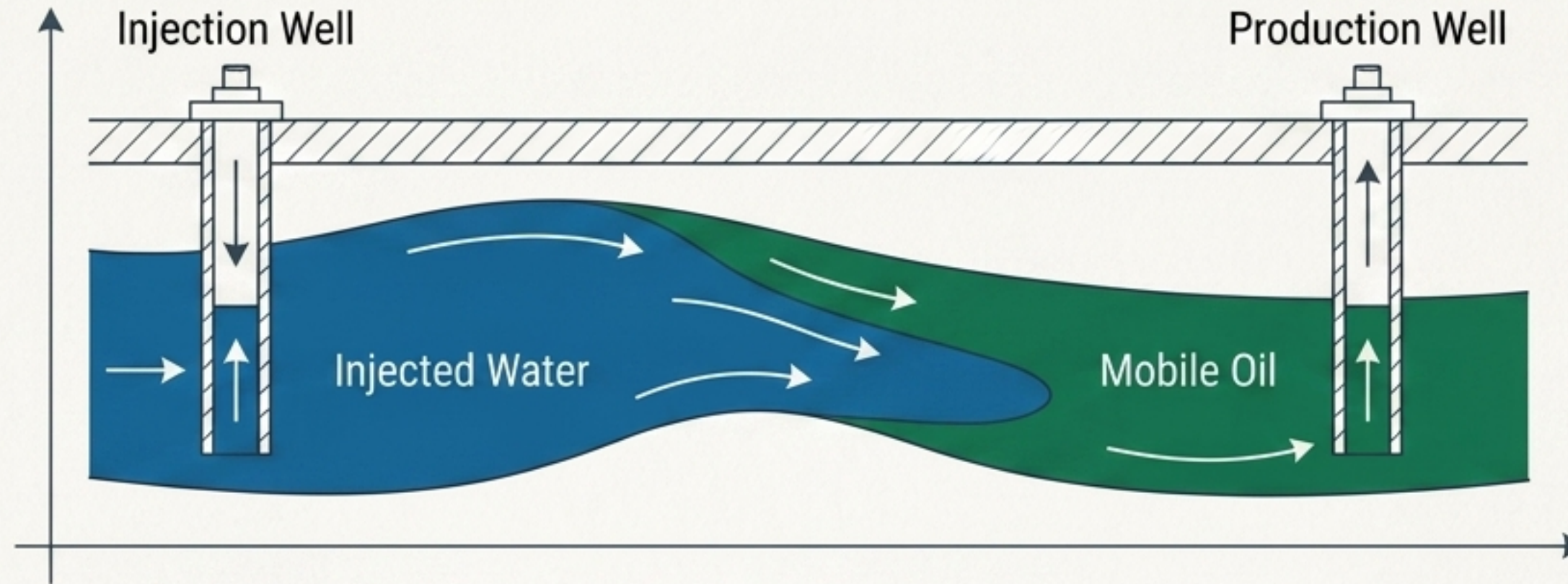


# Enhanced Oil Recovery: The Mechanics of Waterflooding

A conceptual synthesis of fluid dynamics, sweep efficiencies, and reservoir management.



## Core Objectives of Waterflooding

- 1. Increase Recovery Factor:** Maximize swept oil volume.
- 2. Pressure Maintenance:** Fill the voidage left by produced fluids.
- 3. BHP Management:** Maintain reservoir pressure and Bottom Hole Pressure (BHP) above the bubble point pressure.

## Why Water?

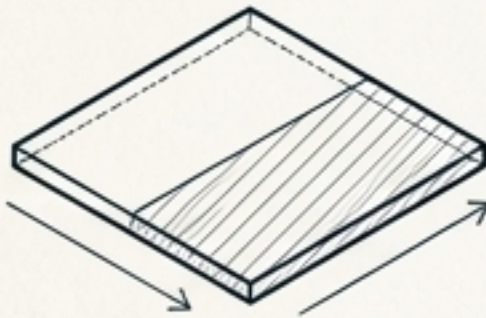
- ✓ High availability in operational regions.
- ✓ Low capital cost relative to other injection fluids.
- ✓ Ease of continuous injection into geological formations.
- ✓ High baseline efficiency in displacing mobile oil.

# The Sweep Efficiency Equation Framework

$$\text{Core Equation: } RF = E = E_a \times E_z \times E_d$$

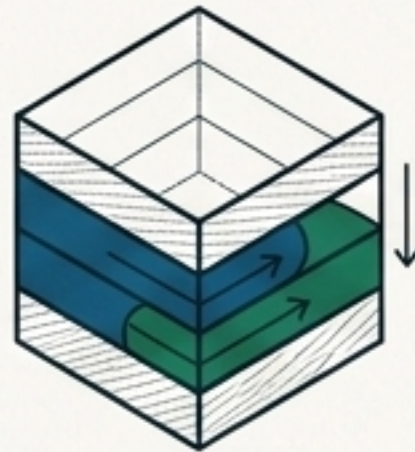
(and  $N_p = N \times E_a \times E_z \times E_d$ )

**$E_a$**   
(Areal Sweep Efficiency)



Fraction of the floodable horizontal pore volume area swept.

**$E_z$**   
(Vertical Sweep Efficiency)



Fraction of the floodable vertical plane swept, driven by stratification.

**$E_d$**   
(Displacement Efficiency)



Microscopic fraction of oil displaced in the invaded zones.

# The Physics Engine: Endpoint Mobility Ratio (M)

**Concept:** M is the principal indicator for sweep efficiency, defining the behavioral relationship between injected water and in-place oil.

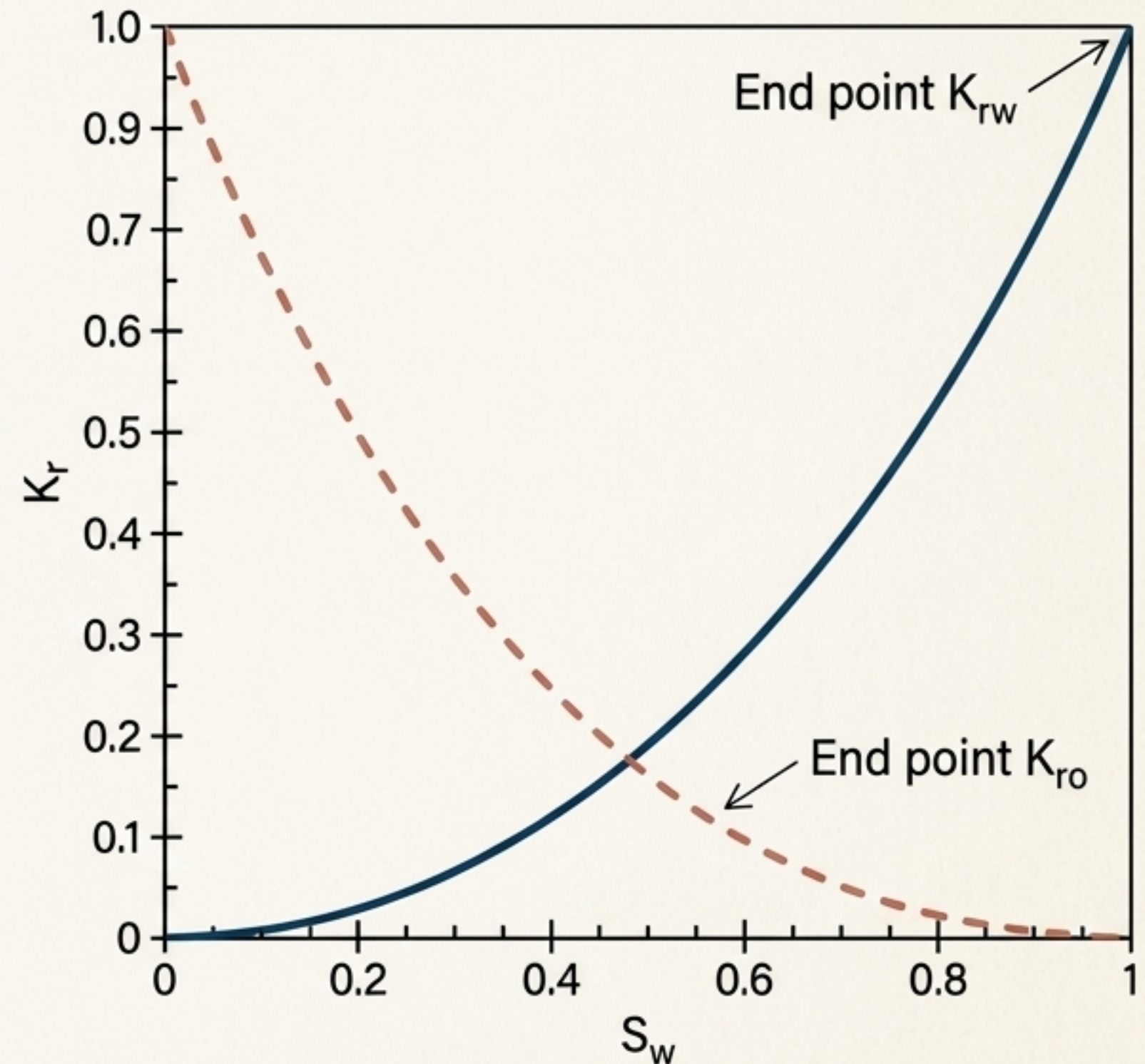
$$M = \frac{k'_{rw} / \mu_w}{k'_{ro} / \mu_o}$$

Where:

$k'_{rw}, k'_{ro}$  = End point relative permeabilities

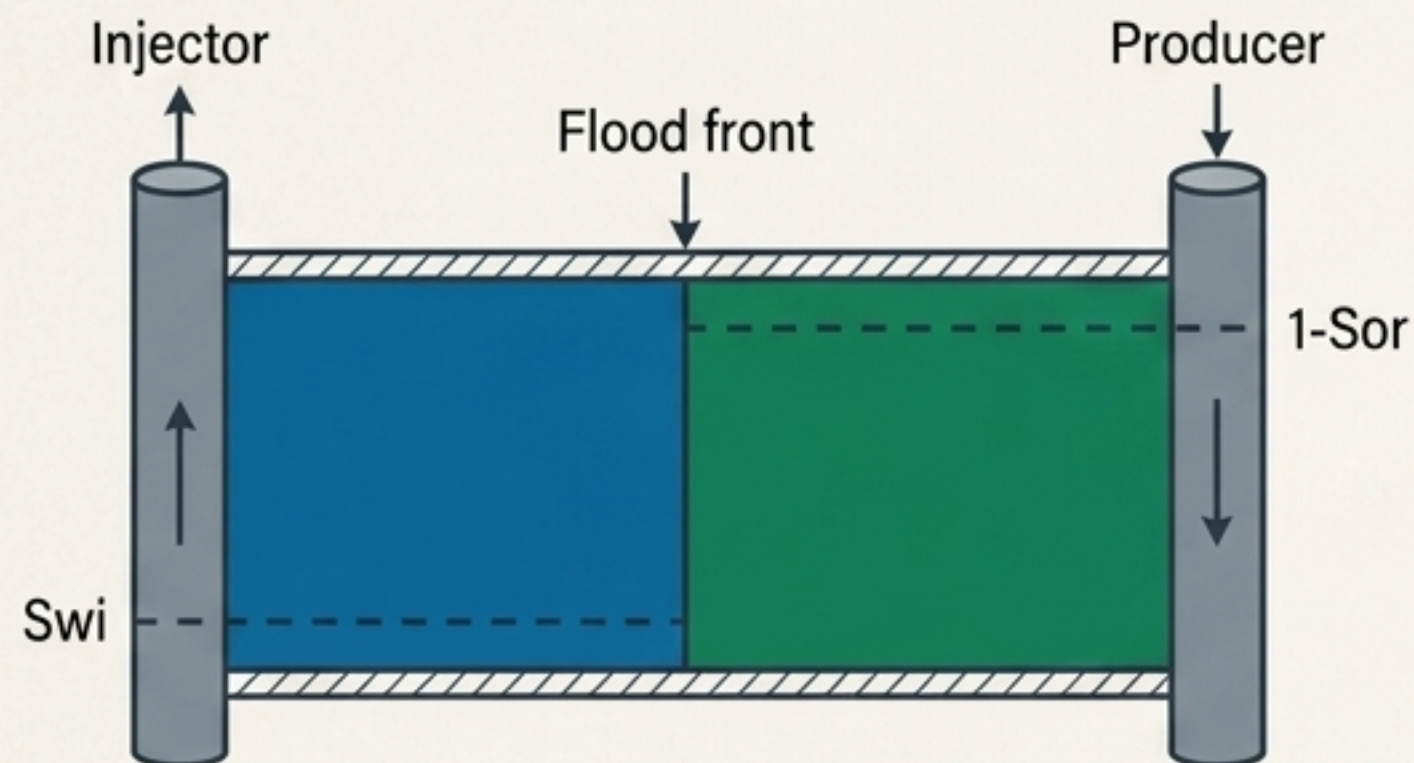
$\mu_w, \mu_o$  = Fluid viscosities

**Key Insight:** Mobility dictates whether displacement operates as an optimal, unified front or fragments into inefficient channeling.



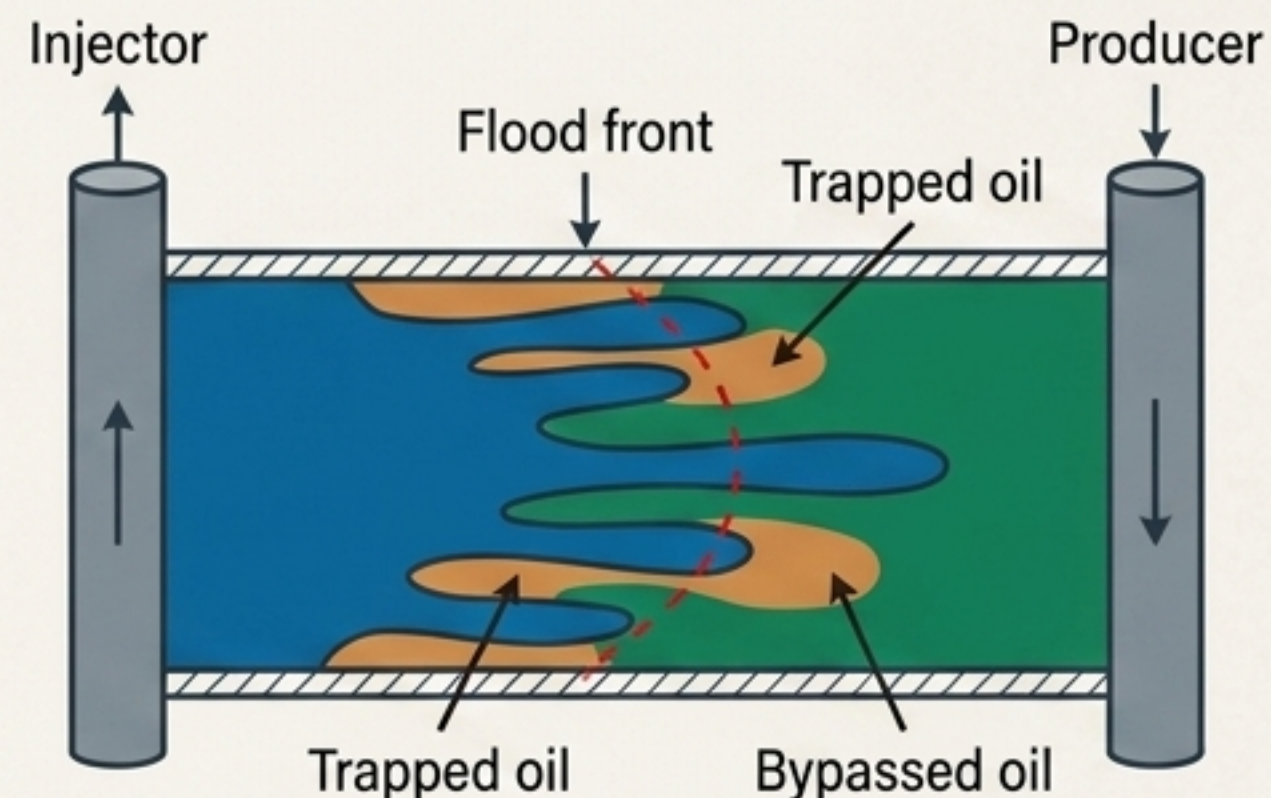
# Diagnostic Matrix: Displacement Dynamics

## Ideal Displacement ( $M \leq 1$ )



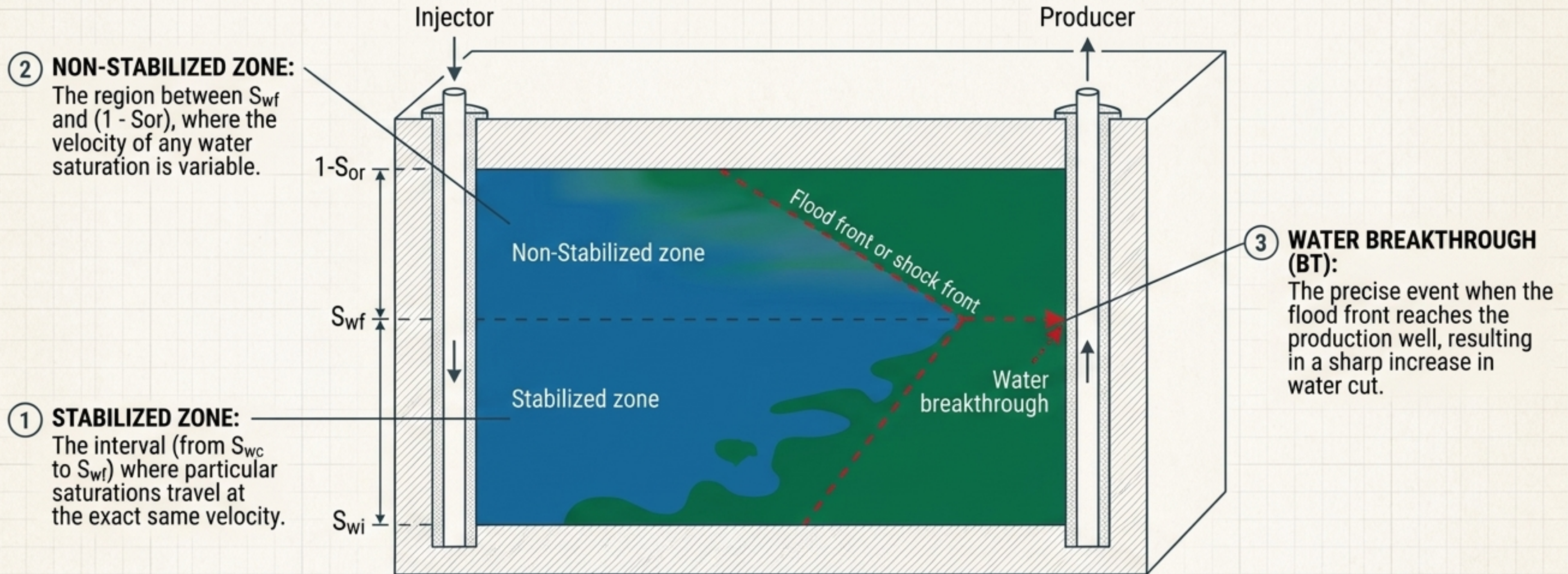
- **Mechanism:** Piston-like fashion. Sharp transition from  $S_{or}$  to maximum oil saturation ( $1 - S_{wi}$ ).
- **Flow:** Oil travels at a velocity equal to or greater than water.
- **Result:** High sweep efficiency, volume of oil recovered matches water injected. Moveable oil volume =  $(1 - S_{or} - S_{wi})V_p$ .

## Non-Ideal Displacement ( $M > 1$ )



- **Mechanism:** Leaky piston-like fashion. Water is more mobile than oil.
- **Flow:** Tongues of water bypass the oil.
- **Result:** Poor saturation profiles, early breakthrough, and extensive bypassed areas. (Most common in nature).

# ANATOMY OF THE FLOOD FRONT



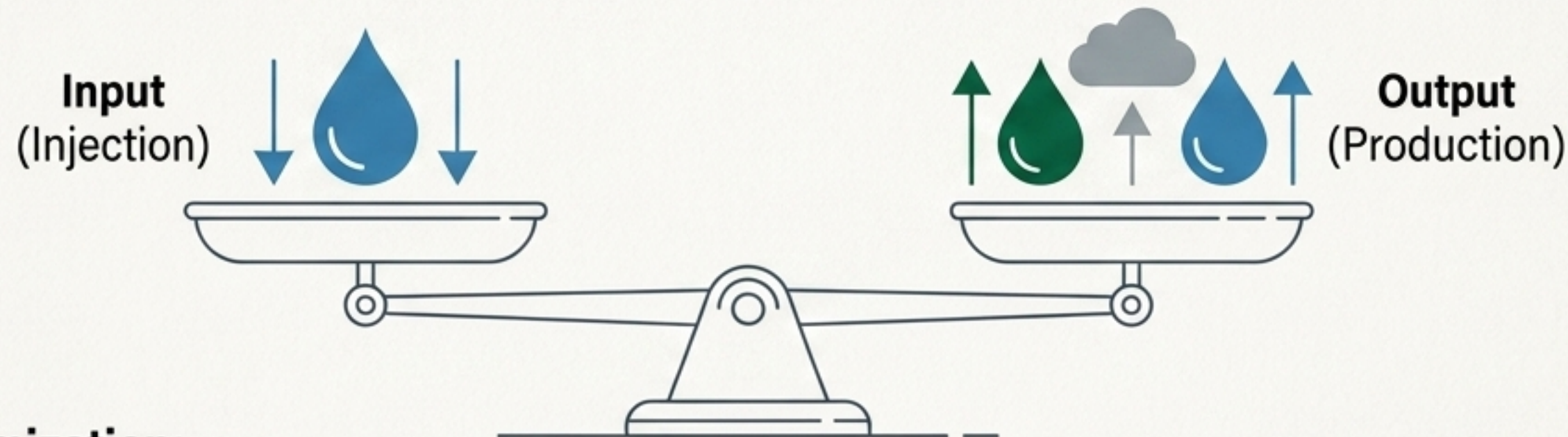
**INSIGHT:** In non-ideal displacement, only a fraction of movable oil is recovered at Breakthrough. Displacing remaining movable oil requires 5 to 6 times its volume in injected water.

# System Balance: Voidage Replacement Ratio (VRR)

**Definition:** The ratio of total injected reservoir volume to total produced reservoir volume at reservoir conditions.

$$\text{VRR} = \frac{Q_i \beta_w}{Q_o \beta_o + ((R_p - R_s) \times Q_o \beta_g) + Q_w \beta_w}$$

**Volume Definitions:** Produced oil volume + Produced Free gas volume + Produced water volume (at reservoir conditions)



## Operational Optimization:

### VRR = 1 (Optimum):

Injected volume perfectly displaces the hydrocarbon volume.

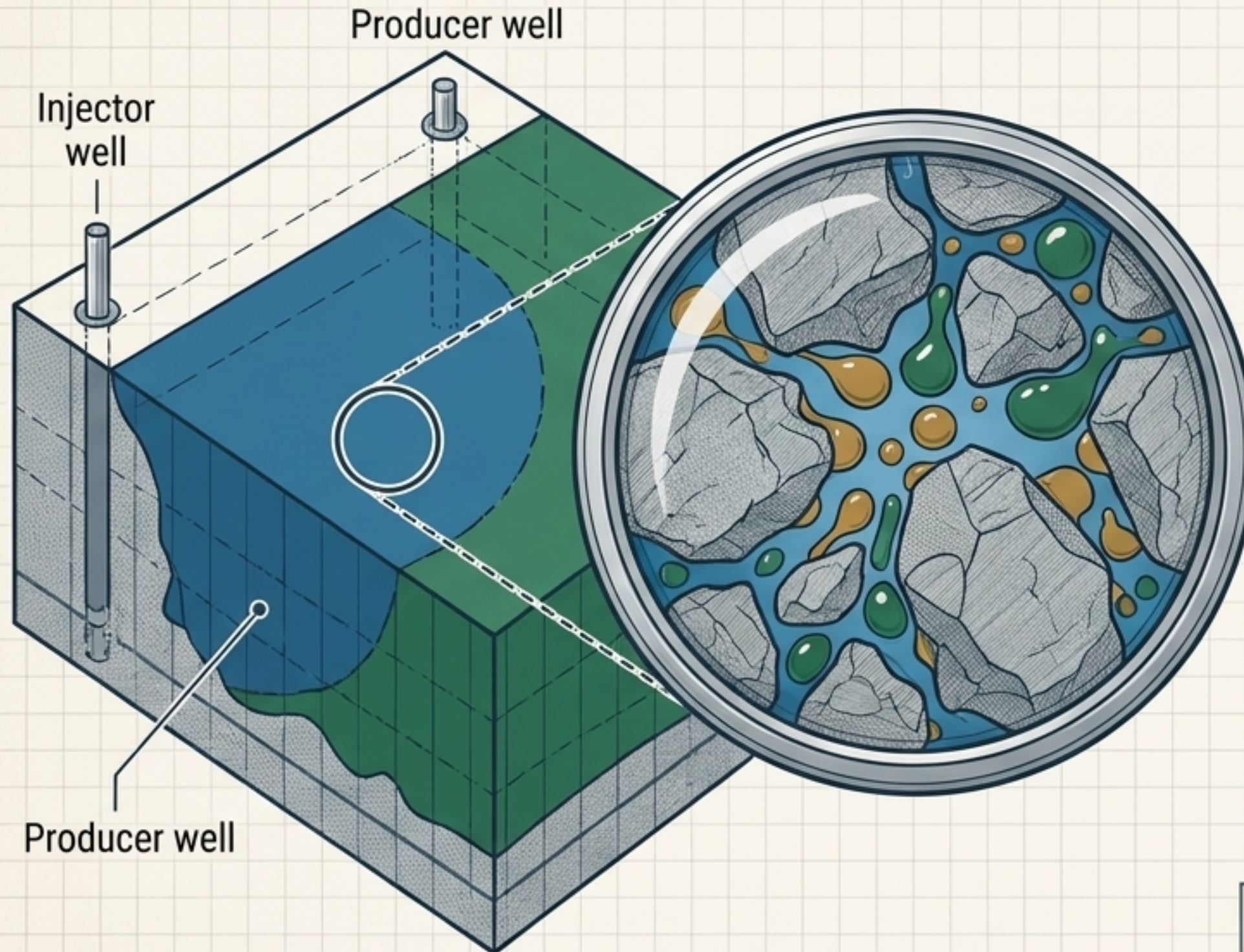
### VRR < 1 (Deficit):

Injection is too low. Results in pressure drop and productivity decline.

### VRR > 1 (Surplus):

Injection is too high. Risks early breakthrough or water fingering.

# Displacement Efficiency ( $E_d$ ): The Pore-Scale Lens



## Definition:

The fraction of oil displaced by injected fluid within the specifically invaded portion of the reservoir.

### The Baseline Equation

$$E_D = \frac{\bar{S}_w - S_{wi}}{1 - S_{wi}} \quad (\text{Assuming no free gas})$$

#### At Breakthrough:

Calculated using the average water saturation at breakthrough

$$E_{DBT} = \frac{\bar{S}_{wBT} - S_{wi}}{1 - S_{wi}}$$

#### After Breakthrough:

Calculated for each selected value over time

$$E_D = \frac{\bar{S}_{w2} - S_{wi}}{1 - S_{wi}}$$

Note: Remaining oil here is trapped at the pore scale, requiring chemical/miscible EOR methods to alter fluid properties.

# Spatial Geometry: The 5-Spot Pattern Strategy

## Primary Recovery Stage:

All wells act as oil producers with arbitrary spacing based on natural pressure.

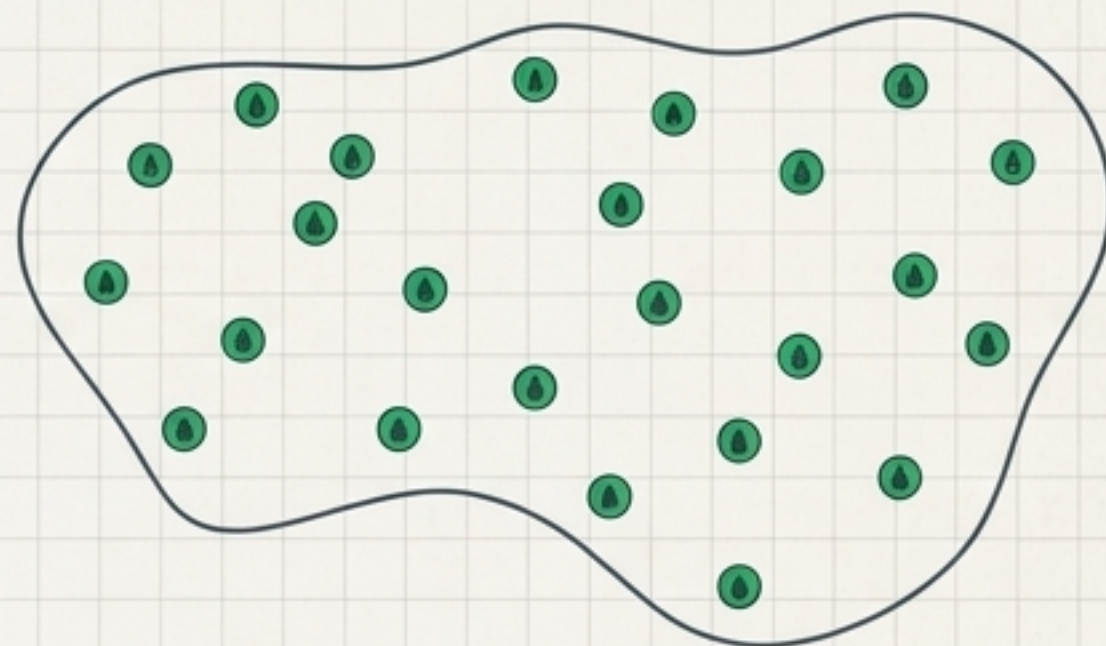
## Secondary Recovery Conversion:

Waterflooding necessitates geometric structuring. Selected producers are converted into injectors.

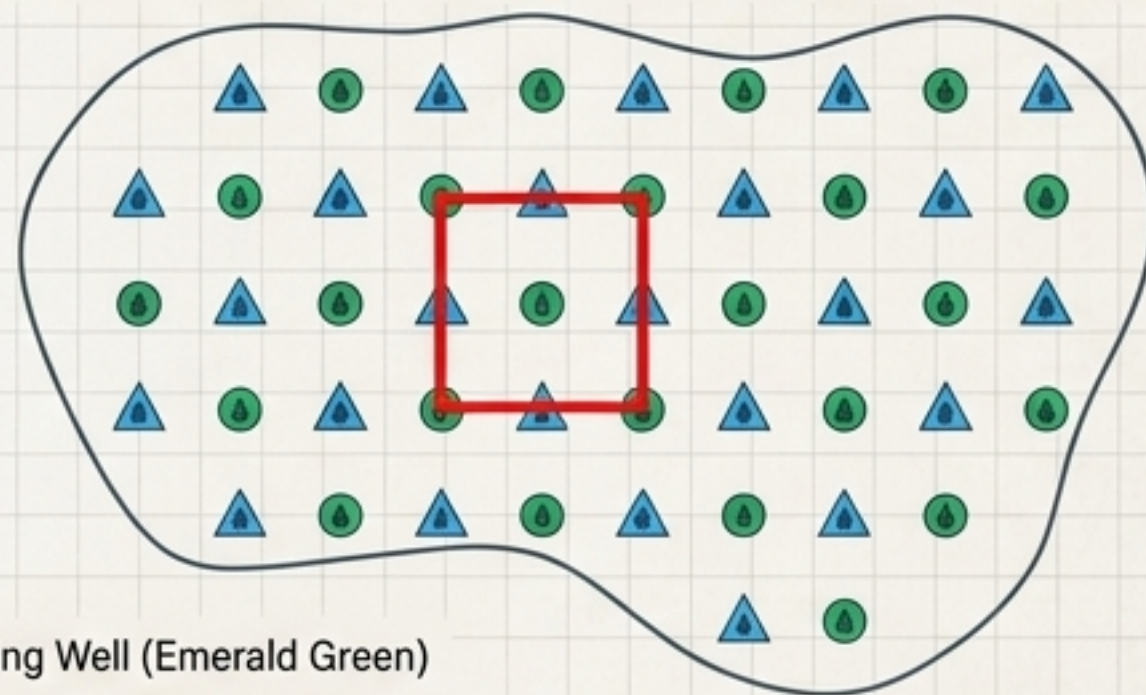
## The 5-Spot Architecture:

A highly symmetrical grid where a central producing well is surrounded by four corner injection wells, optimizing the driving force of the water bank across the horizontal plane.

PRIMARY PRODUCTION STAGE



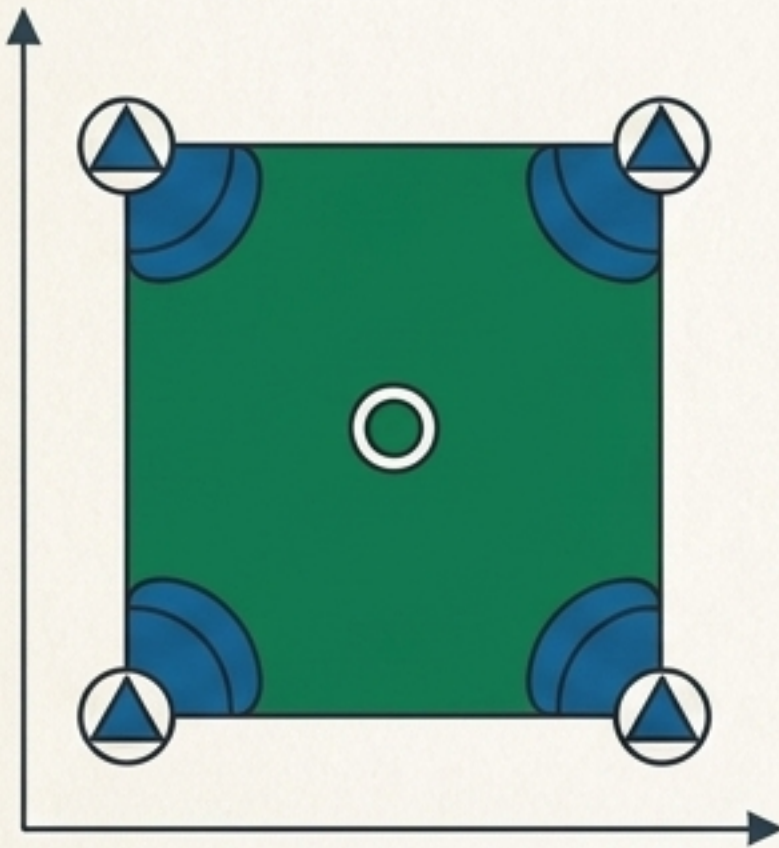
SECONDARY RECOVERY: 5-SPOT CONVERSION



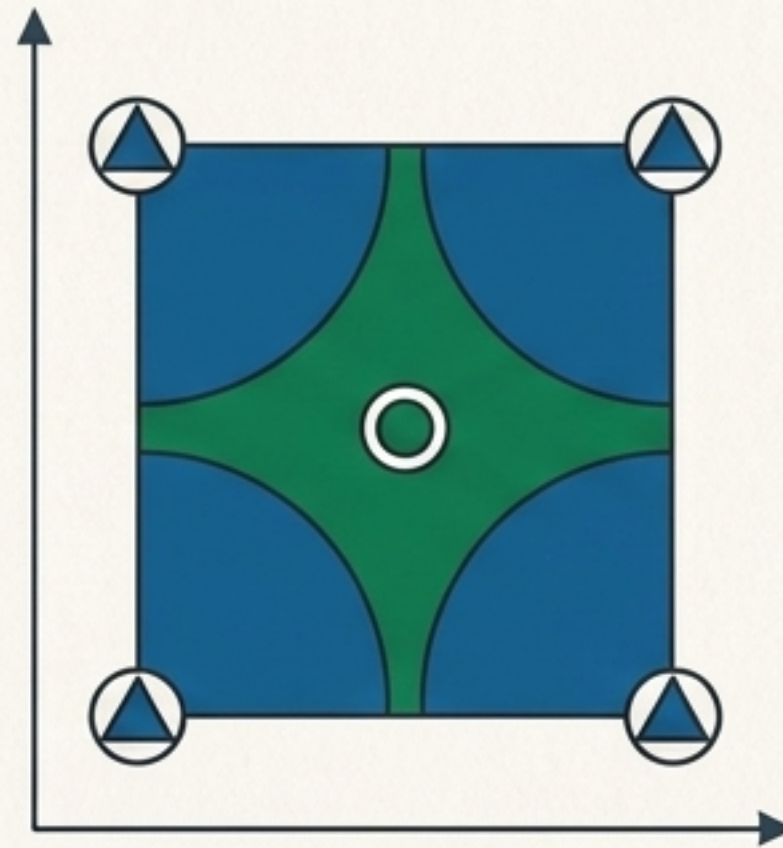
- Producing Well (Emerald Green)
- ▲ Water Injection Well (Marine Blue)

# Areal Sweep ( $E_a$ ): Horizontal Front Progression

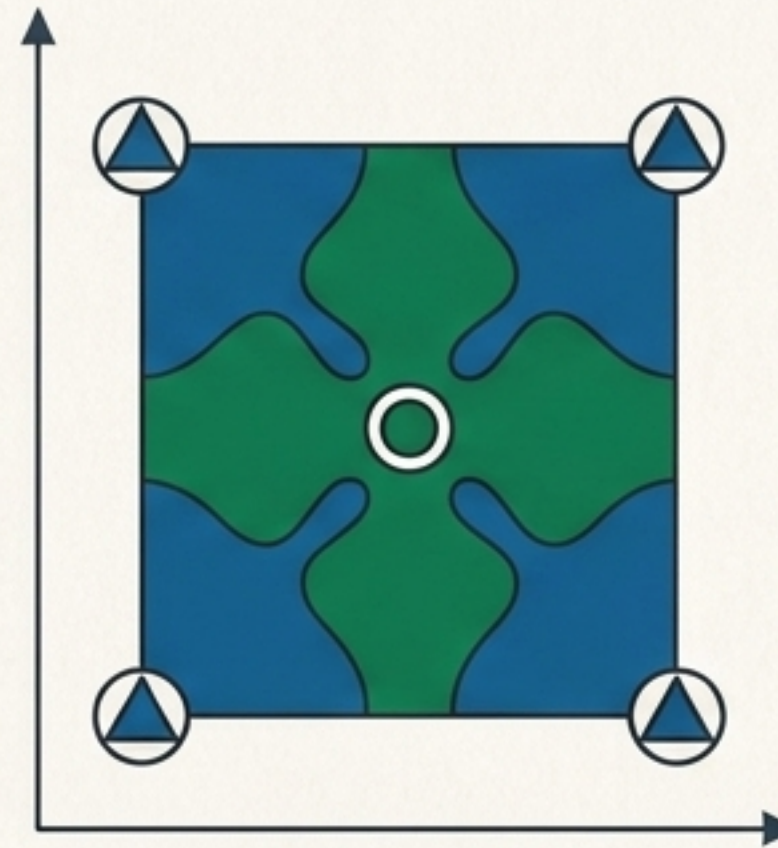
**Definition:** The fraction of the floodable horizontal pore volume area swept by injected fluid.



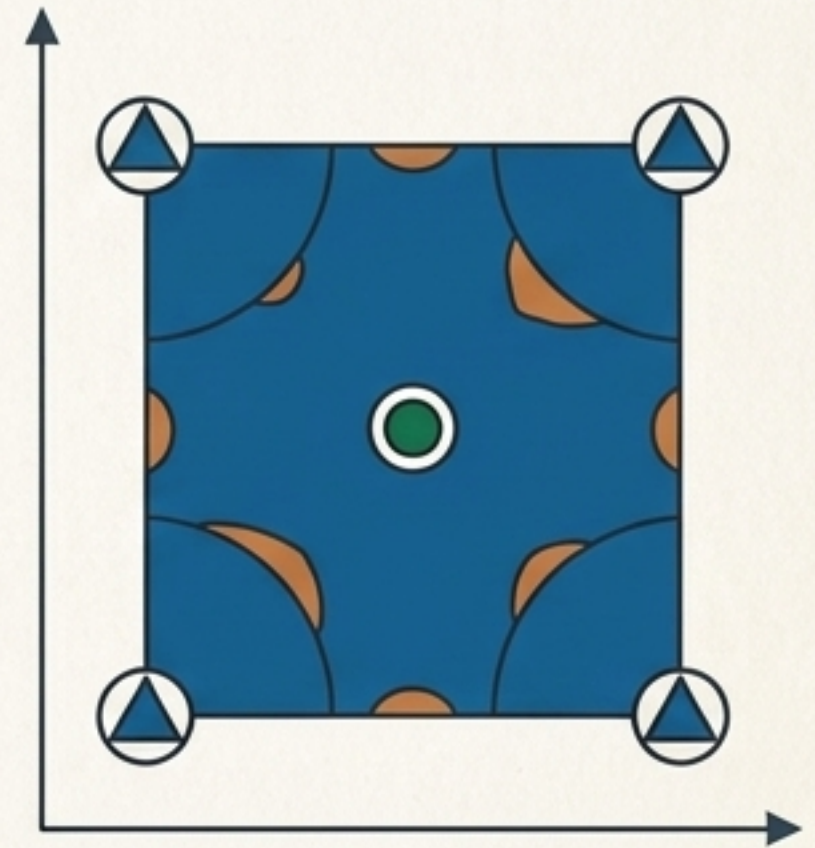
**Phase 1 (Early Time):**  
Water front takes a cylindrical form around injectors. Oil wells produce 100% oil.  $E_a$  increases steadily.



**Phase 2 (Expanding Banks):**  
Continuous injection develops pressure distribution and streamlines. Water banks expand toward the center.



**Phase 3 (Mid-Life / Breakthrough):**  
The water front deforms from its cylindrical shape and cusps into the production well. Oil and water are produced simultaneously.



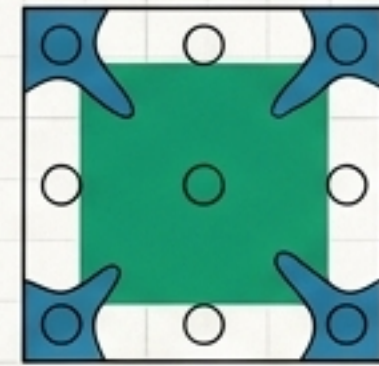
**Phase 4 (Late-Life):**  $E_a$  continues to increase, but at a significantly slower rate. Wells produce primarily water.

# The Impact of Fluid Mobility on Areal Sweep

## The Craig et al. Study:

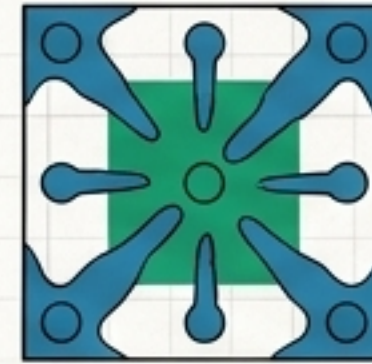
### Unfavorable Mobility (M=1.43):

Rapid fingering and cusping.  
Only 65% of the pattern area is swept at breakthrough.



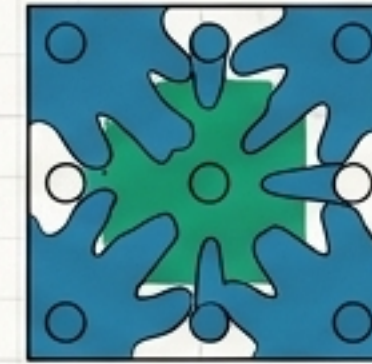
Mobility Ratio=1.43

Water Breakthrough



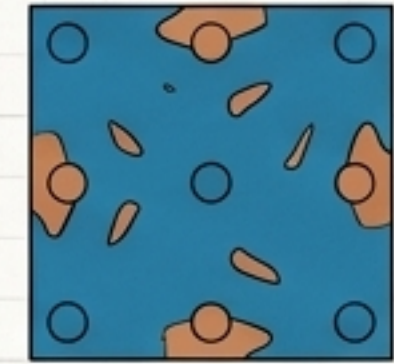
Areal Sweep Efficiency 65%

WOR=0.5



70.5%

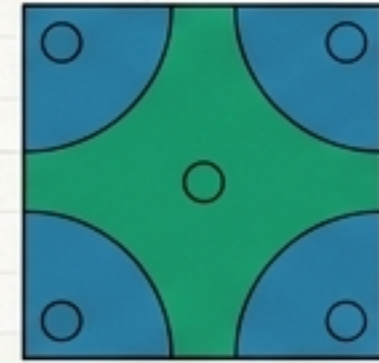
WOR=2



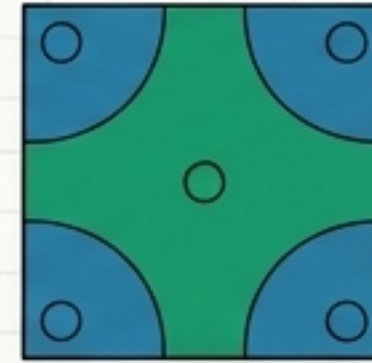
82.2%

### Favorable Mobility (M=0.4):

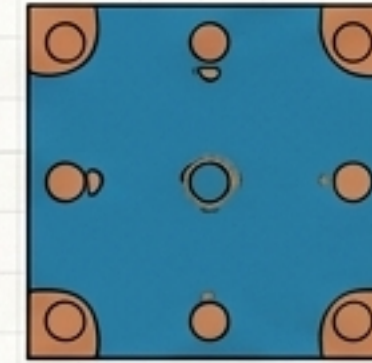
Slower, more uniform front progression. Reaches 82.8% areal sweep efficiency at breakthrough.



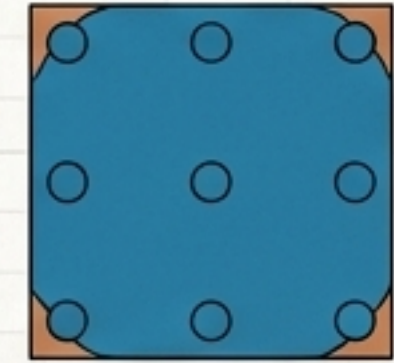
Mobility Ratio=0.4



Areal Sweep Efficiency 82.8%



WOR=0.6  
87.4%

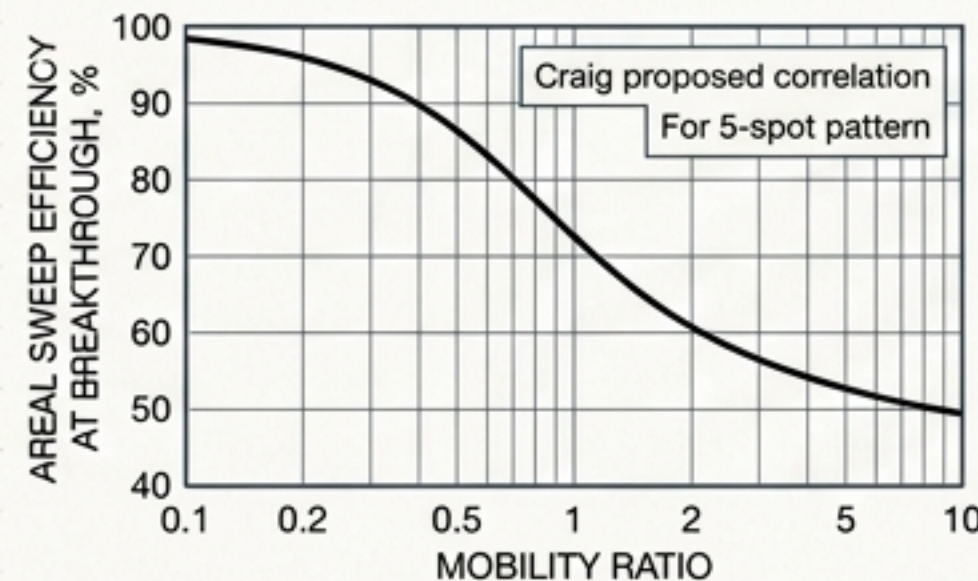


WOR=4.7  
95.6%

## Mathematical Correlations:

### Before Breakthrough:

$$E_A = \frac{W_{inj}}{(PV)(\bar{S}_{wBT} - S_{wi})}$$



### After Breakthrough (Dyes et al.):

$$E_A = E_{ABT} + 0.2749 \ln \left( \frac{W_{inj}}{W_{iBT}} \right)$$

# Vertical Sweep (Ez): Overcoming Stratification

**Definition:** The fraction of the floodable pore volume in the vertical plane swept by the injected fluid.

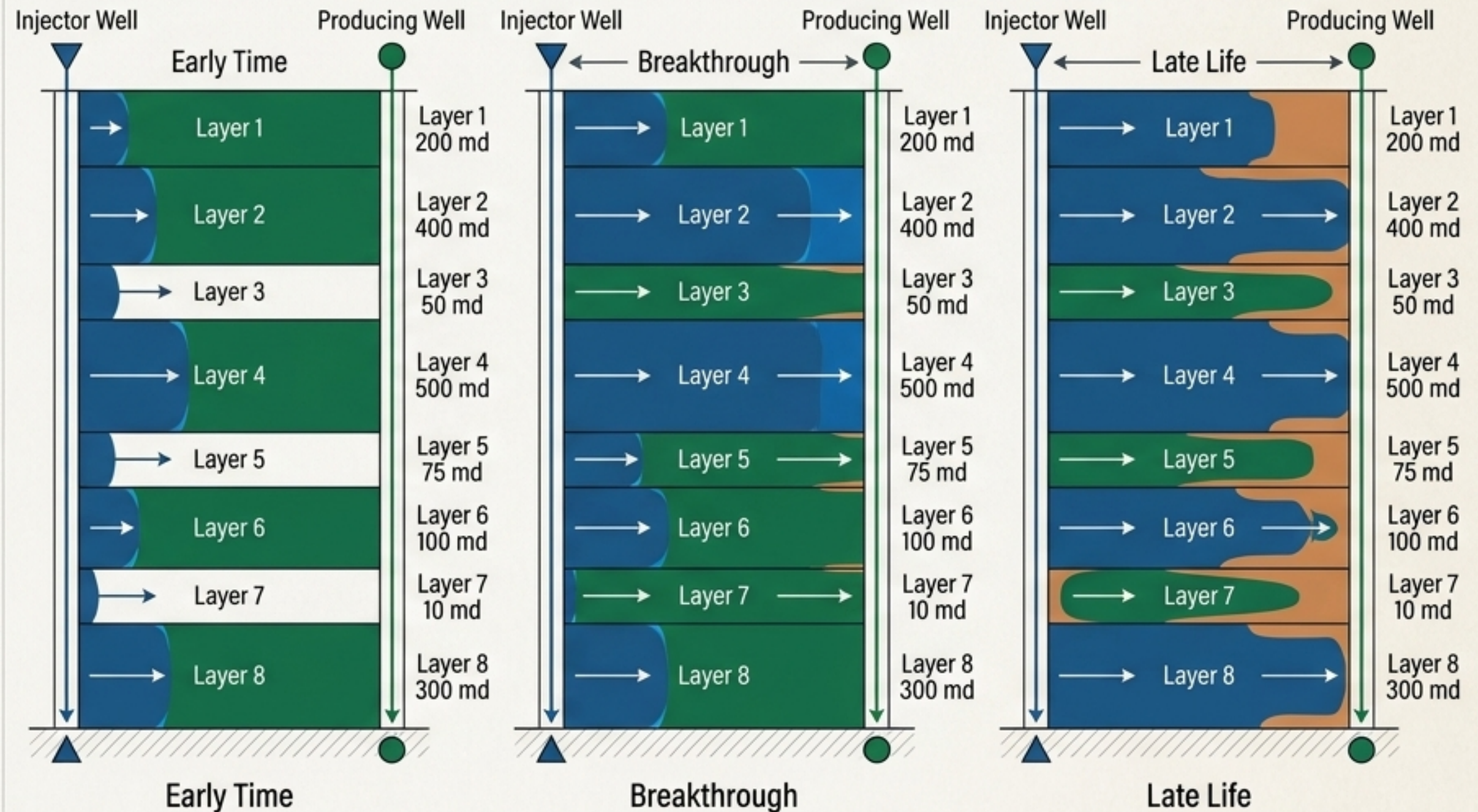
## The Permeability Driver:

Most oil reservoirs encompass multiple geological layers with highly variable horizontal permeabilities.

## The Challenge of Heterogeneity:

- High-permeability layers act as “thief zones,” aggressively pulling injected water and leading to early breakthrough in specific strata.
- Low-permeability layers remain largely unswept, leaving significant hydrocarbon volumes stranded despite prolonged injection.

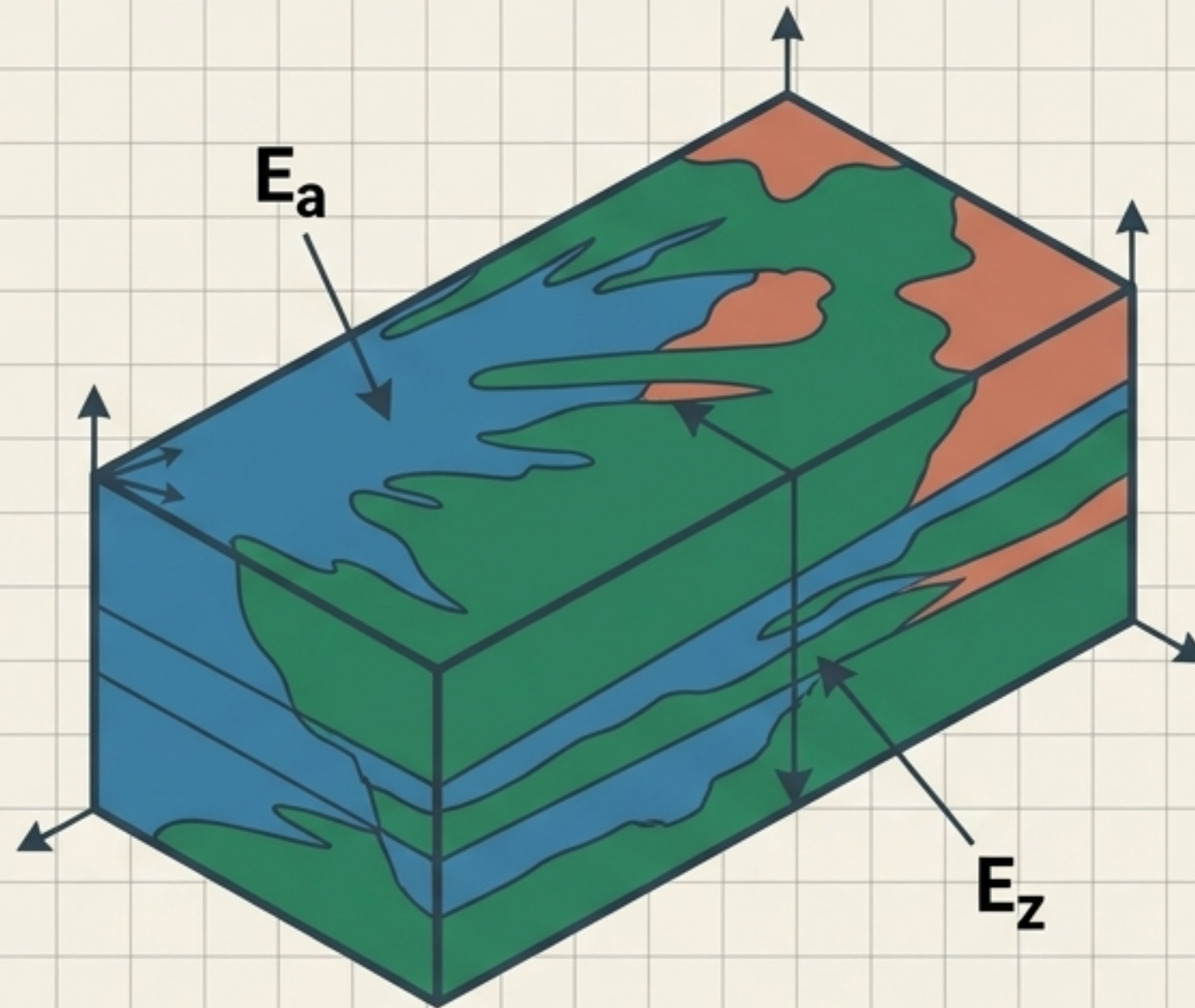
Time-Lapse Vertical Cross-Section: Injected Water Advancing Through Heterogeneous Layers



# Volumetric Sweep ( $E_v$ ): The Macroscopic Block

## Definition:

Also known as Macroscopic displacement efficiency. It accounts for the combined sweep efficiency effect caused by both areal and vertical heterogeneities.



**Key Concept:** A successful flood must push fluids aggressively across the horizontal map ( $E_a$ ) while simultaneously balancing flow regimes across highly stratified geological layers ( $E_z$ ). A failure in either plane drastically reduces  $E_v$ .

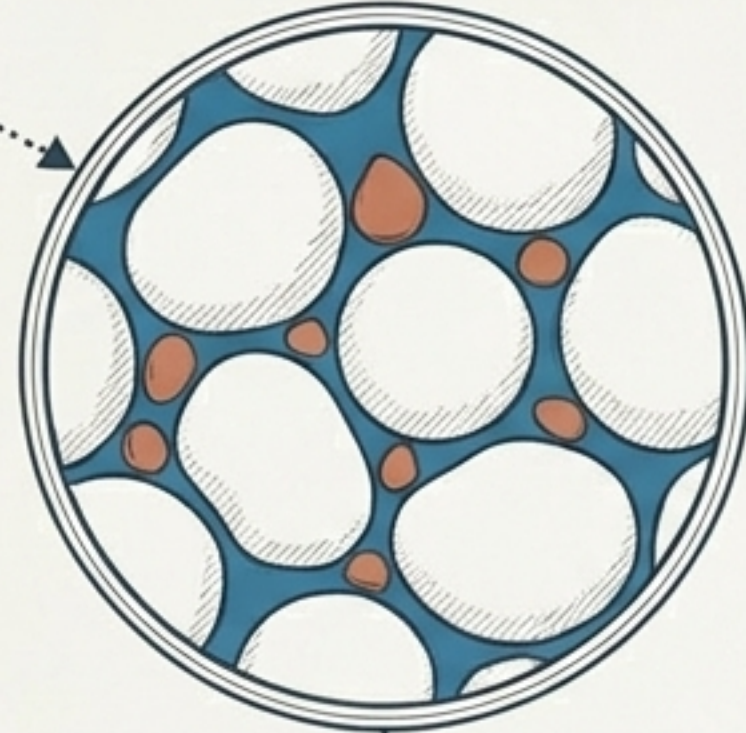
$$\text{Equation: } E_v = E_a \times E_z$$

# The Final Target: Distinguishing Trapped vs. Bypassed Oil

**The Mission of Advanced EOR:** The ultimate target of any intervention is attacking the oil left behind by primary waterflooding. We must precisely categorize this remaining resource:

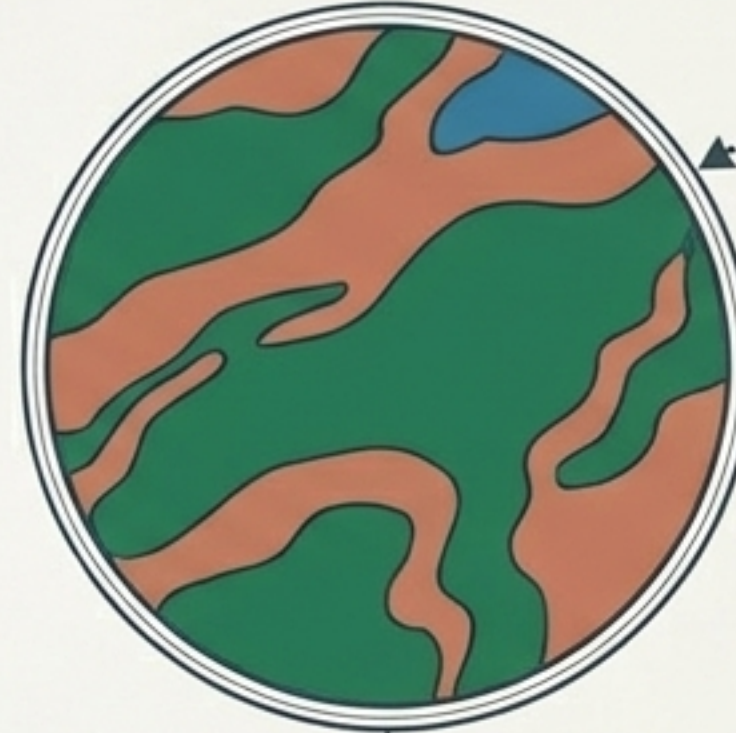
## Category 2: Trapped Oil (The Micro Failure)

- **Driven By:** Poor Displacement Efficiency ( $E_d$ ). Oil is immobilized by capillary forces even within heavily swept zones (10 - 30% saturation).
- **Solution:** Surfactants, polymers, or miscible gas injection to fundamentally alter fluid physics.



## Category 1: Bypassed Oil (The Macro Failure)

- **Driven By:** Poor Volumetric Sweep ( $E_v$ ) due to permeability channeling and geometric un-swept zones (20 - 60% saturation).
- **Solution:** Profile modification, infill drilling, and conformance control.



Injection Well

Production Well

SWEPT area

Un-SWEPT area

Un-SWEPT area